

and the length of channel that gives the maximum rate is not quite the same for lower pressures as it is for the higher pressures; for example, with the 0.1 inch length of channel, at 2 inches orifice pressure, the gas rate is below the maximum point, and at 6 inches the same orifice gives a gas rate which is slightly beyond the maximum rate. In other words, the length of channel which will give the maximum rate is slightly different for different pressures.

The reason for the rapid increase in rate of flow when a short-channel orifice replaces the sharp-edge orifice is that with the short channel the constriction of the gas stream, which occurs in the case of the sharp-edge orifice with gases as well as with liquids, is reduced, and although the maximum velocity may be less, the average velocity at the orifice (and therefore the volume of discharge) is increased, the rate of increase being shown by the rising curves of Fig. 7. At a certain point this effect is a maximum, and as the length of the channel is further increased the friction increases and reduces the flow. The length of tube for maximum flow varies slightly with the pressure, as shown by the curves.

TABLE 1.—Rate of Flow and Discharge Coefficient of Channel Orifices. (See Fig. 7)

[Specific gravity of gas, 1.0; size of orifice, No. 2 drill (0.221 inch diameter); orifices with 45° approach]

Length of channel	At 2-inch pressure		At 4-inch pressure		At 6-inch pressure	
	Rate of flow	Discharge coefficient (<i>K</i>)	Rate of flow	Discharge coefficient (<i>K</i>)	Rate of flow	Discharge coefficient (<i>K</i>)
Inch	Foot ³ /hour		Foot ³ /hour		Foot ³ /hour	
0.00.....	63.6	0.705	89.5	0.703	109.6	0.703
.04.....	68.5	.760	95.1	.747	115.6	.741
.08.....	71.3	.792	108.6	.853	138.1	.886
.10.....	77.1	.855	114.9	.903	140.6	.902
.125.....	81.6	.906	114.6	.901	139.2	.893
.25.....	79.2	.879	111.4	.875	136.4	.875
.50.....	78.2	.868	110.0	.865	135.0	.865
1.00.....	75.0	.832	105.7	.831	129.7	.832

It will be noted in Table 1 that the orifice of 0.08 inch length of channel has a discharge coefficient at 2 inches pressure of 0.792; at 4 inches, 0.853; and at 6 inches, 0.886, although for other lengths the differences due to the variation of pressure are less. It follows, then, that it is impossible to design an orifice of the channel type within the critical length—namely, 0.00 to about 0.125 inch—that will give a perfectly constant discharge coefficient, although